

Practice Quiz 01A — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 01

Truth tables and logical equivalence.

1. Use a complete truth table to determine whether

$$P \wedge (P \vee Q) \quad \text{and} \quad P$$

are logically equivalent. State your conclusion.

Solution

P	Q	$P \vee Q$	$P \wedge (P \vee Q)$
F	F	F	F
F	T	T	F
T	F	T	T
T	T	T	T

The final column agrees with the column for P in every row. Therefore the formulas are logically equivalent. This is the absorption law $P \wedge (P \vee Q) \equiv P$. ■